GenAI Models Explained: GPT, DALL·E, Codex, Stable Diffusion

At a glance

Three architectures power almost every famous GenAI model: **Transformers** (GPT, Codex), **diffusion** (Stable Diffusion, DALL·E 2 and 3), and the older **GANs**. Transformers generate text and code token by token; diffusion models generate images by removing noise; GANs generate images through a contest between two networks.

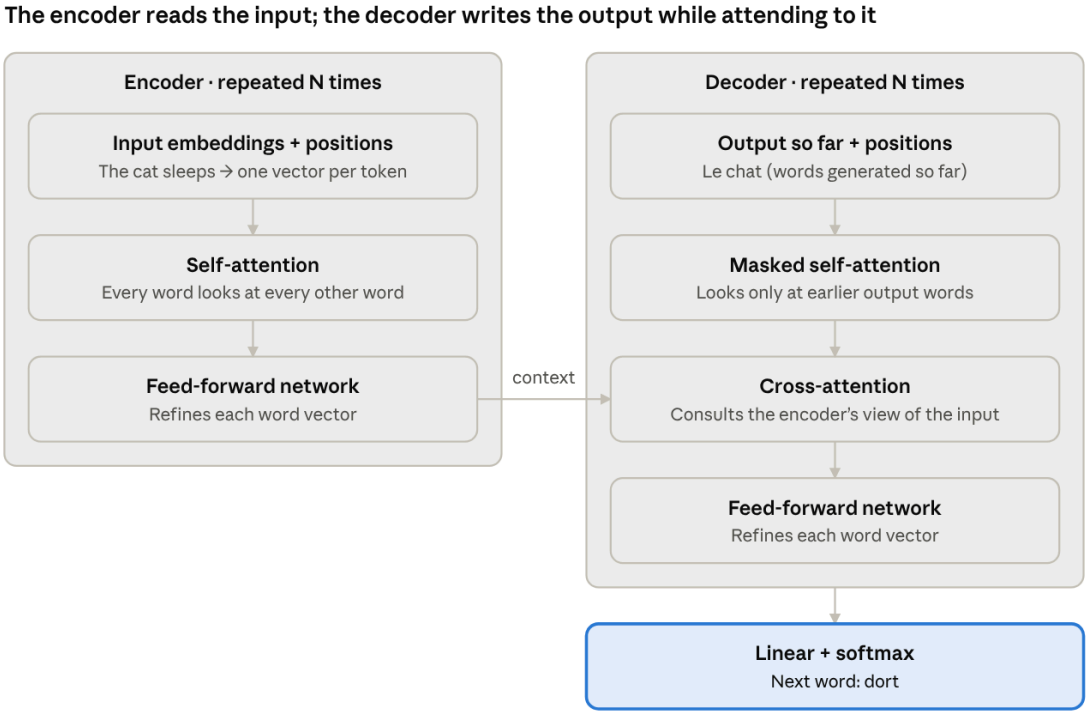
Model	Maker	Generates	Core architecture	Open or closed	First released
GPT	OpenAI	Text (now also images, audio, code)	Decoder-only Transformer	Closed (API and ChatGPT)	2018
Codex	OpenAI	Code; today a full coding agent	GPT Transformer fine-tuned on code	Closed model; open-source CLI	2021
DALL·E	OpenAI	Images from text	Transformer (v1) → diffusion (v2, v3)	Closed	2021
Stable Diffusion	Stability AI	Images from text or images	Latent diffusion (U-Net, later diffusion Transformer)	Open weights	2022

Read Parts 1–4 for how each architecture works, Parts 5–8 for the models themselves, and Part 9 for enterprise use.

Part 1 — What are Transformers?

A Transformer is a neural network that processes a whole sequence at once and uses **attention** to decide which parts of the input matter for each word. Introduced by Google researchers in the 2017 paper "Attention Is All You Need", it replaced older recurrent networks (RNNs, LSTMs) that read one word at a time and forgot long-range context.

The original architecture



Original Transformer · encoder + decoder, translating English to French

The encoder builds an understanding of the full input; the decoder writes the output one word at a time, checking that understanding through cross-attention at every step.

Key components

Component	What it does	Simple analogy
Tokenizer	Splits text into tokens (words or word pieces)	Cutting a sentence into puzzle pieces
Embedding	Turns each token into a vector of numbers	Giving each piece coordinates on a meaning map
Positional encoding	Adds word-order information	Numbering the pieces so order isn't lost
Self-attention	Each token weighs every other token's relevance	Reading a sentence and linking "it" to the noun it refers to
Multi-head attention	Many attention calculations in parallel	Several readers each tracking a different detail

Component	What it does	Simple analogy
Feed-forward layer	Processes each token's vector further	Each reader thinking over their notes
Residual + normalization	Keeps training stable across many layers	Guard rails on a long staircase
Softmax output	Turns scores into probabilities for the next token	Picking the most likely next word

Three families of Transformers

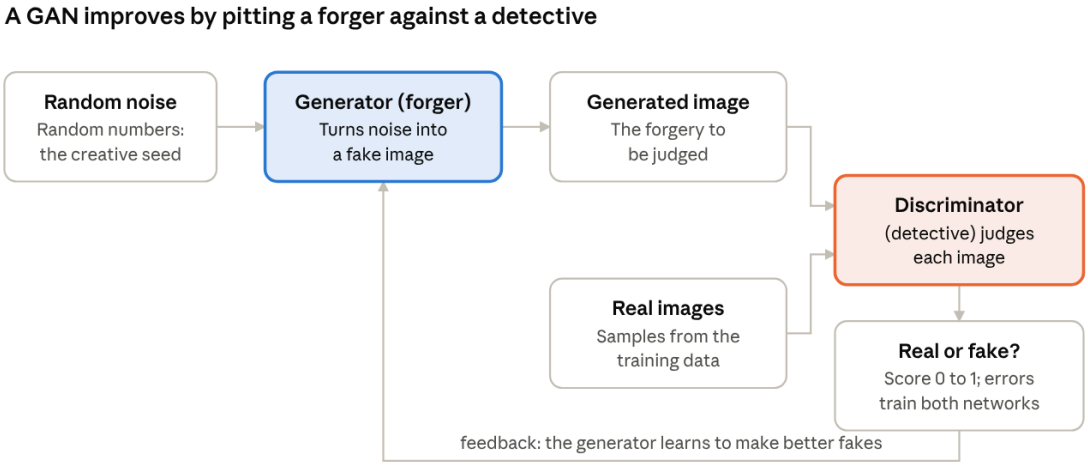
Family	Uses	Good at	Examples
Encoder-only	Encoder stack	Understanding: search, classification, embeddings	BERT, RoBERTa
Decoder-only	Decoder stack (no cross-attention)	Generation: chat, writing, code	GPT, Codex, Claude, Llama
Encoder-decoder	Both	Input-to-output tasks: translation, summarisation	T5, BART

Transformers are no longer just for text. **Vision Transformers** split images into patches treated like tokens, and **diffusion Transformers (DiT)** now power image and video generators such as Stable Diffusion 3 and Sora.

Why they won: they train in parallel on GPUs, capture long-range context, and keep improving as models and data scale up.

Part 2 — What are GANs?

A **Generative Adversarial Network (GAN)** is two neural networks trained against each other: a *generator* that creates fake data and a *discriminator* that tries to tell fake from real. Invented by Ian Goodfellow and colleagues in 2014, GANs produced the first convincingly realistic AI-generated faces.



GAN training loop · generator vs discriminator

Think of a forger and an art detective. Each time the detective catches a fake, the forger learns what gave it away. Training ends when the detective can only guess at 50/50 — the fakes are indistinguishable from real images.

How training works, step by step

1. Feed random noise into the generator to produce a batch of fake images.
2. Mix them with a batch of real images.
3. The discriminator labels each image real or fake; its mistakes update the discriminator's weights.
4. The generator is updated to make the discriminator call its fakes "real".
5. Repeat thousands of times until both networks stop improving.

Strengths and weaknesses

Strengths	Weaknesses
Very fast generation: one pass through the generator	Unstable training: the two networks can fall out of balance
Sharp, high-detail images	**Mode collapse**: the generator learns to produce only a few kinds of output
Good for narrow, well-defined domains (faces, textures)	Harder to control with text prompts than diffusion models

Famous GANs and uses

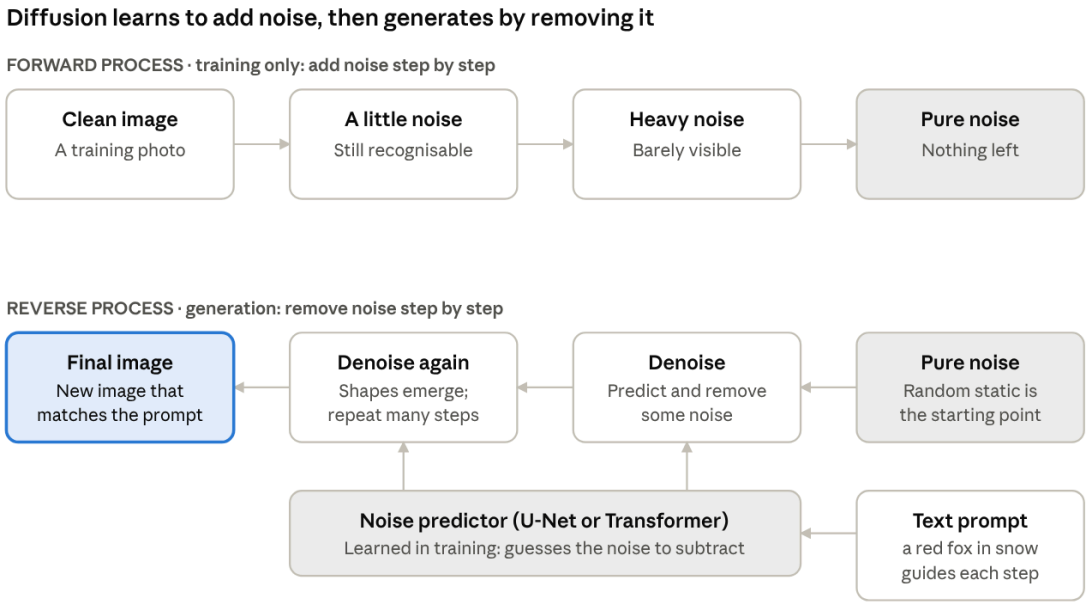
GAN	Known for	Enterprise use
StyleGAN (NVIDIA)	Photorealistic human faces	Synthetic faces for testing, avatars

GAN	Known for	Enterprise use
CycleGAN	Style transfer without paired examples (horse ↔ zebra)	Converting photo styles; medical image translation
ESRGAN	Super-resolution (upscaling)	Enhancing low-resolution product or satellite images
Tabular GANs (CTGAN)	Realistic synthetic table data	Privacy-safe synthetic customer and transaction data

Where GANs stand today: diffusion models have overtaken GANs for general text-to-image generation, but GANs remain useful where speed matters, for upscaling, and for synthetic data.

Part 3 — What is diffusion?

A **diffusion model** creates images by starting from random noise and removing it a little at a time until a clear picture appears. It learns how to do this by watching real images being gradually destroyed by noise and practising how to undo each step. Diffusion became the standard for image generation after the 2020 DDPM paper and now powers Stable Diffusion, DALL·E 2 and 3, Midjourney, and video models like Sora.



Diffusion · forward noising in training, reverse denoising in generation

The forward process is only used to create training examples. At generation time the model runs the reverse process alone, and the text prompt steers every denoising step toward the image you asked for.

Key ideas in plain words

Concept	What it means
Forward process	Add small amounts of Gaussian noise over many steps until the image is pure static
Reverse process	Starting from static, repeatedly predict and remove noise
Noise predictor	The neural network (a U-Net or a diffusion Transformer) trained to guess the noise in an image
Text conditioning	A text encoder (such as CLIP or T5) turns the prompt into vectors the predictor uses as guidance
Classifier-free guidance (CFG)	A dial for how strictly to follow the prompt: higher = more literal, lower = more creative
Sampling steps	How many denoising steps to run; fewer = faster, more = finer detail
Latent diffusion	Denoise a compressed version of the image instead of every pixel, making it much faster (Part 8)

What diffusion can do beyond text-to-image

- **Image-to-image** — start from a sketch or photo with some noise added, and redraw it in a new style.
- **Inpainting** — regenerate only a masked area, such as removing an object or changing clothing.
- **Outpainting** — extend an image beyond its original borders.
- **Video, audio and 3D** — the same principle applied over frames, sound waves or 3D shapes.

Why diffusion beat GANs for images: training is stable, outputs are more diverse, and text prompts give fine control. The trade-off is speed: many denoising steps per image, though newer techniques cut this to a handful.

Part 4 — Transformers vs GANs vs diffusion

Use Transformers for language and code, diffusion for images and video, and GANs where speed or synthetic data matters most.

Aspect	Transformer (autoregressive)	GAN	Diffusion
Core idea	Predict the next token from all previous ones	Generator and discriminator compete	Learn to reverse gradual noising

Aspect	Transformer (autoregressive)	GAN	Diffusion
Main outputs	Text, code, also images and audio	Images, synthetic data	Images, video, audio
Generation speed	Medium: one token at a time	Fast: a single pass	Slower: many denoising steps
Training stability	Stable	Unstable, needs careful tuning	Stable
Output diversity	High	Can suffer mode collapse	High
Prompt control	Excellent	Limited	Excellent
Famous examples	GPT, Codex, Claude, Gemini, Llama	StyleGAN, CycleGAN	Stable Diffusion, DALL·E 3, Midjourney, Sora
Year introduced	2017	2014	2015 (idea); 2020 (DDPM breakthrough)

The lines are blurring: diffusion models now use Transformers as their noise predictor, and multimodal GPT models generate images directly.

Part 5 — GPT (Generative Pre-trained Transformer)

GPT is OpenAI's family of decoder-only Transformers: models pretrained on vast text to predict the next token, then fine-tuned to follow instructions. The name says it all — **Generative** (it writes), **Pre-trained** (on internet-scale data first), **Transformer** (the architecture from Part 1).

How GPT works

1. **Pretraining** — read trillions of tokens of web pages, books and code; learn to predict each next token.
2. **Instruction tuning (SFT)** — train on example prompts with ideal answers.
3. **RLHF** — humans rank answers; the model learns to prefer helpful, safe responses.
4. **Reasoning training** (from the o-series onward) — reward correct answers to checkable problems so the model learns to think step by step before replying.
5. **Inference** — your prompt is tokenized, run through the Transformer stack, and the reply is generated one token at a time.

Version history

Release	Model	What changed
Sep 2026	GPT-6 Astra	New flagship aimed at long-running agentic and research work (reported)

Release	Model	What changed
Jul 2026	GPT-5.6 (Sol, Terra, Luna)	Three tiers — flagship, balanced, fast/low-cost — launched with ChatGPT Work and Codex
Apr 2026	GPT-5.5	Better coding, computer use and research
Aug 2025	GPT-5	Unified fast and reasoning modes in one system
Sep 2024	o1 (o-series)	First "reasoning" models that think before answering
May 2024	GPT-4o	Natively multimodal: text, images and voice in one model
Mar 2023	GPT-4	Big jump in reasoning; accepts image input
Nov 2022	GPT-3.5 / ChatGPT	Chat interface plus RLHF brought GenAI to the public
2020	GPT-3 (175B parameters)	Few-shot learning from examples in the prompt
2019	GPT-2 (1.5B)	Coherent multi-paragraph text
2018	GPT-1 (117M)	Proved pretrain-then-fine-tune works

What GPT models can do

Capability	Example
Write and edit	Emails, reports, marketing copy, translations
Summarise and extract	Turn a 50-page contract into key clauses and dates
Answer and reason	Explain concepts, solve math, analyse data
See and hear	Read charts, invoices and screenshots; hold voice conversations
Generate images	GPT image models now create images directly inside ChatGPT
Act	Call tools and APIs, browse, run code as an agent

Access: through ChatGPT (consumer and enterprise plans), the OpenAI API, and cloud partners such as Microsoft Azure OpenAI.

Part 6 — Codex (GPT for code)

"Codex" names two different things: a 2021 GPT model fine-tuned on code that powered the first GitHub Copilot (now retired), and OpenAI's current **Codex coding agent**, launched in 2025, which writes, tests and fixes code on its own.

How it works

Code is just another language to a Transformer. Codex-style models are GPT models trained further on billions of lines of public code, so they learn syntax, libraries and common patterns. Given a comment like `# function that checks if a number is prime`, the model predicts the code that usually follows.

The modern Codex agent adds the agent loop: it reads your repository, plans a change, edits files, runs tests in a sandbox, reads the errors, and repeats until tests pass — then proposes a pull request for human review.

Evolution

Date	Milestone
Jul 2026	Codex app merged with the ChatGPT desktop app; GPT-5.6 available in Codex (Wikipedia)
Apr 2026	"Codex for (almost) everything" update adds computer use, an in-app browser, memory and 90+ plugins (review)
Mar 2026	Codex Security agent for finding and fixing vulnerabilities; GPT-5.4 for Codex
Feb 2026	Codex desktop app (macOS, then Windows) for running many agents in parallel
May 2025	Codex cloud agent in ChatGPT: each task runs in its own sandboxed cloud computer (TechCrunch)
Apr 2025	Codex CLI: open-source coding agent for the terminal
2023	Original Codex model retired; code ability folded into GPT-3.5 and GPT-4
Aug 2021	Codex model released via API; powers GitHub Copilot (from June 2021 preview)

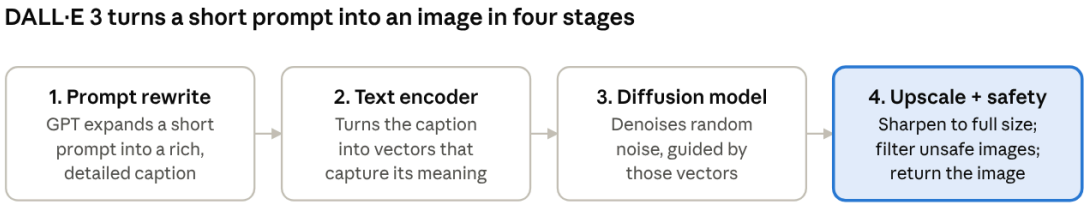
What it's used for

Task	Example
Code generation	Build a REST endpoint from a description
Bug fixing	Read a failing test, find the cause, patch it
Code review	Comment on a pull request's logic and style
Refactoring and migration	Upgrade a codebase from one framework version to another
Tests and docs	Write unit tests and README files for existing code
Security	Scan commits for vulnerabilities and propose fixes

Comparable tools include GitHub Copilot, Anthropic's Claude Code, Cursor and Google's Gemini Code Assist.

Part 7 — DALL·E (text to image)

DALL·E is OpenAI's text-to-image model series: type a description, get an original image. The name blends the artist Salvador Dalí with Pixar's robot WALL·E. Its architecture changed completely between versions — a Transformer in v1, diffusion in v2 and v3.



DALL·E 3 text-to-image pipeline · 4 stages

DALL·E 3's key innovation was step 1: because it was trained on highly detailed captions and a GPT model rewrites your prompt in that style, it follows long, specific instructions far better than earlier models.

Versions

Version	Year	Architecture	Notable for
GPT image models	2025 onward	Image generation built natively into multimodal GPT	Accurate text in images, editing through conversation; now OpenAI's main image generator
DALL·E 3	2023	Diffusion + GPT prompt rewriting, integrated into ChatGPT	Strong prompt following; readable text in images
DALL·E 2	2022	"unCLIP": CLIP embeddings + diffusion decoder	Photorealism, inpainting, image variations
DALL·E 1	2021	12B-parameter Transformer generating image tokens	Proved text-to-image generation was possible

CLIP: the bridge between words and pictures

CLIP (Contrastive Language–Image Pre-training, 2021) was trained on hundreds of millions of image–caption pairs to place matching images and text close together in the same vector space. This lets a model measure how well an image fits a prompt — a building block used by DALL·E 2, Stable Diffusion and many others.

Enterprise uses

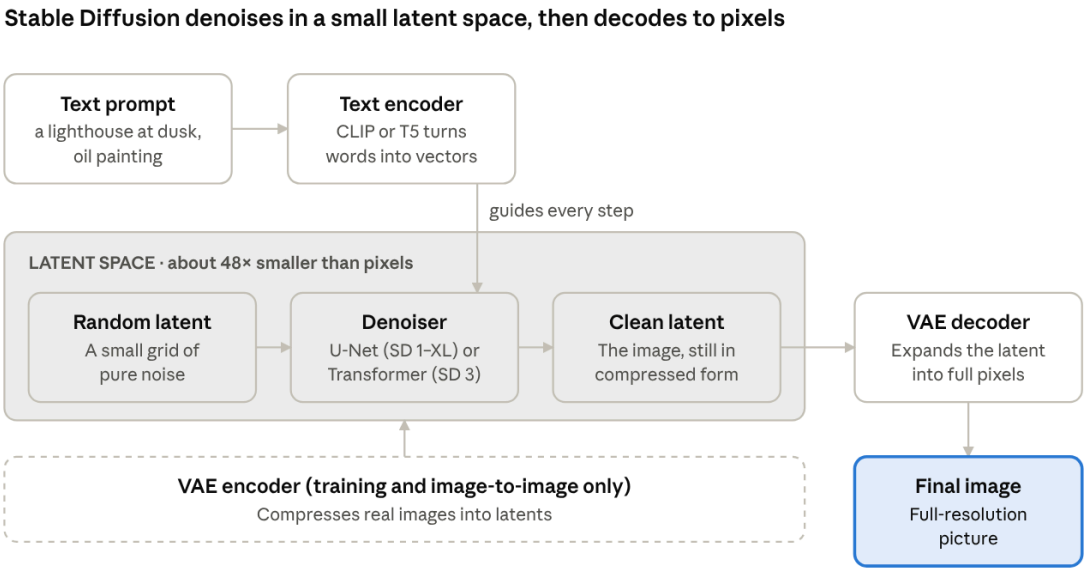
- Marketing visuals and ad variants
- Product concept art and packaging mock-ups
- Presentation and training illustrations
- Storyboards for video and campaigns

OpenAI applies safety filters (for example, blocking images of real public figures by name and imitation of living artists' styles) and attaches C2PA provenance metadata to mark images as AI-generated.

Part 8 — Stable Diffusion

Stable Diffusion is an **open-weight** text-to-image model from Stability AI, first released in August 2022. Because anyone can download and run it, it spawned a huge ecosystem of custom models, plugins and tools — and it is the go-to choice when a company needs image generation on its own infrastructure.

Its big idea is **latent diffusion**: instead of denoising millions of pixels, it denoises a small compressed representation of the image, which makes generation fast enough to run on a consumer GPU.



Stable Diffusion architecture · latent diffusion with text conditioning

Only the dashed VAE encoder is skipped in plain text-to-image generation; it is used to train the model and to start from an existing picture.

Versions

Version	Release	Highlights
SD 3.5 (Large, Large Turbo, Medium)	Oct 2024	Current official family alongside Stable Image Ultra; no SD 4 announced as of mid-2026 (source)

Version	Release	Highlights
SD 3	2024	Switched to a diffusion Transformer (MMDiT); much better text in images
SDXL 1.0	Jul 2023	3.5B-parameter base model plus refiner; 1024×1024 native resolution (source)
SD 2.x	Nov 2022	New text encoder, 768×768 images
SD 1.5	Oct 2022	The most widely adopted open image model; huge fine-tune ecosystem
SD 1.4	Aug 2022	First public release of open weights

The customisation ecosystem

Tool	What it does	Business example
Fine-tuning / DreamBooth	Teaches the model a new subject or style from a few images	Generate images of your own product
LoRA	Small, cheap add-on weights for a style or concept	A brand-style LoRA for consistent marketing art
ControlNet	Controls composition with a pose, edge map or depth map	Keep a product's exact shape while changing backgrounds
Inpainting / outpainting	Edit or extend parts of an image	Remove objects; resize ads for different formats
Negative prompts	List what to avoid	"blurry, watermark, extra fingers"
ComfyUI, Automatic1111	Visual tools for building image workflows	Repeatable design pipelines

Stable Diffusion vs DALL·E

	Stable Diffusion	DALL·E / GPT image
Access	Download weights; run anywhere	API or ChatGPT only
Customisation	Full: fine-tune, LoRA, ControlNet	Limited to prompting and editing
Data privacy	Can run fully on-premises	Data goes to OpenAI's cloud
Ease of use	Needs setup and GPU	Ready to use

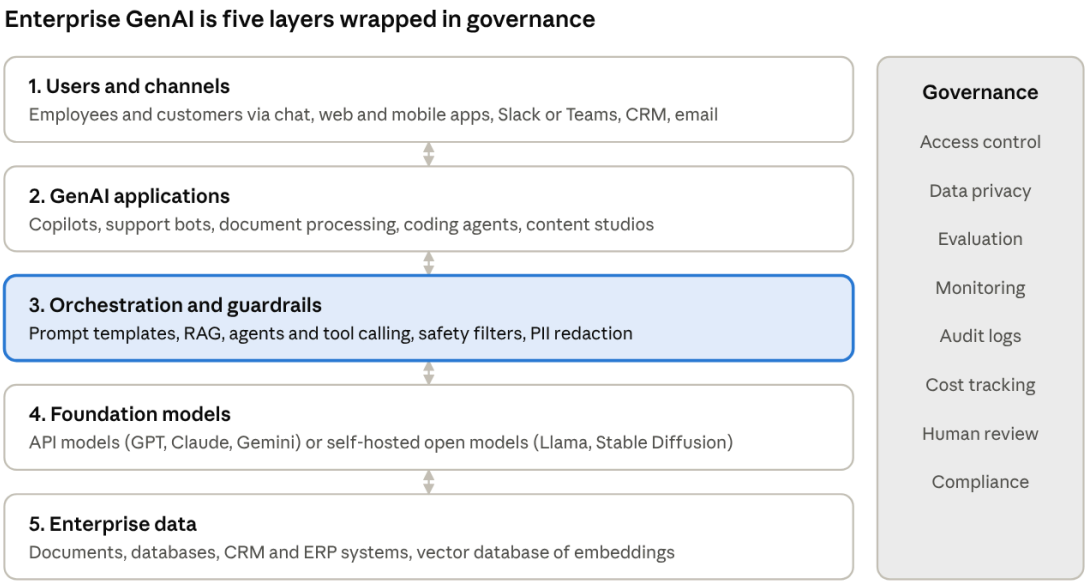
	Stable Diffusion	DALL·E / GPT image
Licensing	Check each model's licence for commercial use	Covered by OpenAI terms

Strong competitors in 2026 include FLUX (Black Forest Labs), Midjourney, Google's Imagen and Adobe Firefly.

Part 9 — Applications in enterprise AI

Enterprises use GenAI mostly to speed up knowledge work: answering questions from company documents, drafting content, writing code, and automating multi-step processes with agents. The model is only one layer; most of the work goes into connecting it safely to company data and workflows.

Enterprise reference architecture



Enterprise GenAI reference architecture · 5 layers + governance

The highlighted orchestration layer is where enterprise value is built: it decides what data the model sees, which tools it can use, and what it is allowed to output.

Use cases by business function

Function	Use case	Model type used	Typical benefit
Customer service	Support chatbot answering from help articles and order data	GPT-class LLM + RAG	Faster responses; agents focus on complex cases

Function	Use case	Model type used	Typical benefit
IT and engineering	Coding agents writing features, tests and fixes	Codex, Claude Code, Copilot	Faster delivery; less routine work
Marketing	Campaign copy, product images, ad variants	GPT + DALL·E / Stable Diffusion	More content, faster A/B testing
Sales	Personalised outreach emails, call summaries, CRM updates	LLM + CRM integration	More time selling
HR	Policy Q&A bot, job descriptions, onboarding guides	LLM + RAG	Fewer repetitive HR tickets
Finance	Invoice extraction, report summaries, variance commentary	Multimodal LLM	Less manual data entry
Legal and compliance	Contract review, clause comparison, regulation summaries	LLM + RAG	Faster first-pass review
Operations	Agents that triage tickets, update systems, draft reports	LLM agents with tools	End-to-end task automation
Data and analytics	Ask questions in plain English, get SQL and charts	Code-generating LLM	Self-service analytics
Product and design	UI mock-ups, concept art, synthetic training images	Diffusion models, GANs	Faster iteration

Which model family for which enterprise need

Need	Best fit	Why
Answer questions from internal documents	Transformer LLM + RAG	Grounded, citable answers
Automate software work	Codex-style coding agents	Read, write and test code
Generate on-brand images	Stable Diffusion fine-tuned with LoRA	Custom style, self-hosted, data stays private
Quick creative images, no setup	DALL·E / GPT image	Ready to use, high quality
Synthetic data for testing or privacy	GANs or LLMs	Realistic data without real customer records
Upscale or enhance images	GANs (e.g. ESRGAN)	Fast, single-pass

Build vs buy

Option	Example	Pros	Cons
Buy a SaaS assistant	ChatGPT Enterprise, Microsoft 365 Copilot, Claude for Work	Fast rollout, no infrastructure	Less customisation
Build on APIs	OpenAI, Anthropic, Google, Azure OpenAI, AWS Bedrock	Custom workflows, your data via RAG	Engineering effort; usage costs
Self-host open models	Llama, Mistral, Stable Diffusion on your own GPUs	Full control and privacy	Highest operational effort

Adoption roadmap

1. **Identify** high-volume, text-heavy tasks with measurable outcomes (tickets, documents, code).
2. **Pilot** with one team, a small use case, and human review of every output.
3. **Ground** the model in company data using RAG; set access permissions.
4. **Evaluate** on a test set of real examples: accuracy, safety, cost per task.
5. **Scale** with governance: monitoring, audit logs, usage policies and training for staff.
6. **Automate** further with agents once trust is established, keeping approval steps for irreversible actions.

Enterprise risks and controls

Risk	Control
Hallucinated answers	RAG with citations; human review for high-stakes outputs
Data leakage	Enterprise agreements with no training on your data; PII redaction; on-prem models for sensitive data
Prompt injection via documents or web pages	Treat retrieved content as data; restrict tool permissions; approval gates
Copyright and brand risk in images	Licensed models; review before publishing; provenance labels
Regulation (EU AI Act, India DPDP Act, sector rules)	Risk classification, documentation, audit trails
Runaway costs	Usage quotas, smaller models for simple tasks, caching

Measuring value

Track time saved per task, resolution or completion rate, quality scores from reviewers, error rate, adoption (weekly active users), and cost per task compared with the manual process.

Quick revision

Term	One-line summary
Transformer	Neural network that uses attention to process a whole sequence at once; basis of GPT and Codex
GAN	Generator and discriminator compete until fakes look real
Diffusion	Learn to undo noise; generate by denoising random static, guided by a prompt
Latent diffusion	Diffusion on a compressed image representation — the trick behind Stable Diffusion's speed
GPT	OpenAI's decoder-only Transformer LLM family for text, reasoning and multimodal tasks
Codex	GPT for code — first an autocomplete model (2021), now an autonomous coding agent
DALL·E	OpenAI's text-to-image models; diffusion-based from v2 onward
Stable Diffusion	Open-weight latent diffusion image model you can run and customise yourself
CLIP	Model that links images and text in one vector space

Self-check questions

1. Why can a Transformer train faster than an RNN?
2. What is the difference between encoder-only and decoder-only Transformers, and which one is GPT?
3. In a GAN, what does the discriminator do, and what is mode collapse?
4. In diffusion, what happens in the forward process and the reverse process?
5. Why does Stable Diffusion work in latent space instead of pixel space?
6. What role does GPT play in DALL·E 3's pipeline?
7. How is today's Codex different from the 2021 Codex model?
8. Which model would you choose to generate on-brand images while keeping data on-premises, and why?
9. Name the five layers of an enterprise GenAI architecture.
10. Give three controls for reducing enterprise GenAI risk.

Sources

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- Stable Diffusion 4 status — DonWeb